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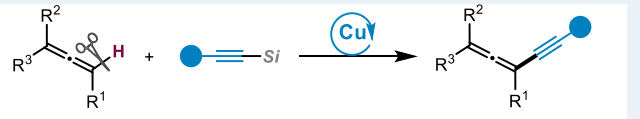
Copper-Catalyzed Site-Selective C(sp^2)−H Alkynylation of Allenes

Xiu Duan,[#] Qihao Jin,[#] Kaifeng Wang,[#] Yifan Sun, Youhao Wei, Zheng Wang, Jiang-Kai Qiu,* Kai Guo,* Xiaoguang Bao,* and Xinxin Wu*

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ABSTRACT: Conjugated allenynes serve as prominently featured structural units in various natural products. However, the absence of an efficient and adaptable synthesis has hindered their broader application in synthetic endeavors and biological investigations. In terms of atom- and step-economy, there is an urgent need for an efficient strategy with a broader substrate scope compared to previous reports. Herein, we report a copper-catalyzed allenic C(sp^2)−H alkynylation of an array of allenes with exceptional site selectivity. While the alkynylation of aliphatic C(sp^3)−H bonds has been extensively studied, radical-mediated regioselective alkynylation of allenyl C(sp^2)−H bonds via direct hydrogen atom transfer (HAT) from the C(sp^2)−H bond of allenes remains an unresolved challenge in synthetic chemistry. This difficulty arises from inherent issues such as the addition of radical intermediates to allenes or alkynes and the regioselectivity of the HAT process. Our protocol addresses these challenges and demonstrates a broader substrate scope, tolerance toward various functional groups, including active sp^3 C−H bonds such as benzyl groups and ethers, and results in a high diversity of products.



KEYWORDS: *hydrogen atom transfer, C(sp^2)−H bond functionalization, copper catalysis, alkynylation, allenynes*

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INTRODUCTION

Conjugated allenynes are notably prominent structural units in various natural products, underscoring their potential significance in organic synthesis (Figure 1A).[1] Nevertheless, the absence of an efficient and adaptable synthesis has impeded their broader application in synthetic endeavors and biological investigations. Consequently, significant efforts have been directed toward developing novel methodologies for the efficient synthesis of allenynes over the past few decades (Figure 1B).[2] However, several issues still need to be addressed in the development of a general strategy for synthesizing allenynes: 1) eliminating the need for laborious substrate synthesis processes; 2) expanding the substrate scope to enable the synthesis of various allenynes; 3) reducing reaction susceptibility; 4) enhancing reaction atom- and step-economy.

Alkynes represent ubiquitous functional groups with high synthetic versatility, often found in drugs, natural products, and organic materials.[3] Recent advancements in radical chemistry have propelled the chemistry of alkynes forward, sparking considerable interest in methods for selectively introducing alkynes into organic molecules.[4] Direct alkynylation of aliphatic C(sp^3)−H bonds has emerged as a popular and efficient approach for synthesizing substituted alkynes (Figure 1C).[5] However, the direct alkynylation of more challenging C(sp^2)−H bonds remains relatively underdeveloped (Figure 1D).

Direct insertion of a functional group into a molecule through C−H bond functionalization stands as one of the most extensively investigated approaches in organic synthetic chemistry.[6] Given the prevalence of C−H bonds in organic molecules, this method offers access to desired products with improved atom- and step-economy compared to classical synthetic methods. Consequently, C−H bond functionalization serves as a potent tool for enhancing synthetic efficiency, particularly in medicinal chemistry, natural product synthesis, and the production of bioactive compounds. To tackle this formidable task, one of the most promising concepts involves the homolytic cleavage of C−H bonds via hydrogen atom transfer (HAT) processes.[7] Significant progress has been achieved in the functionalization of aliphatic C(sp^3)−H bonds through radical HAT processes. However, only a few examples of allenyl C(sp^2)−H functionalization have been reported, typically involving sequential C=C addition/C−H cleavage or concerted C=C activation/C−H deprotonation followed by S_E2'.[8] Direct sp^2 C−H alkenylation of allenes via palladium catalysis utilizing a directing-group strategy was reported by Carreira and coworkers.[9] Additionally, Wang, Li, Zhang et al. reported elegant iron- or cobalt-catalyzed allenic C−H addition to *N,O*-acetals or aldehydes.[10] Recently, independ-

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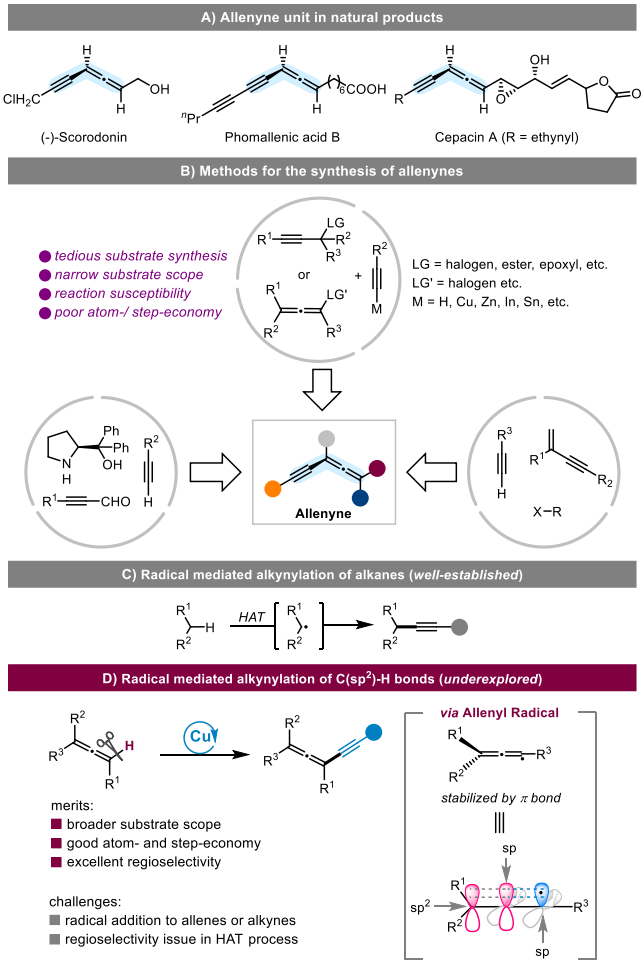


Figure 1. A) Allenyne unit in natural products. B) Methods for the synthesis of allenynes. C) Radical mediated alkynylation of alkanes. D) Radical mediated alkynylation of C(sp^2)−H bonds.

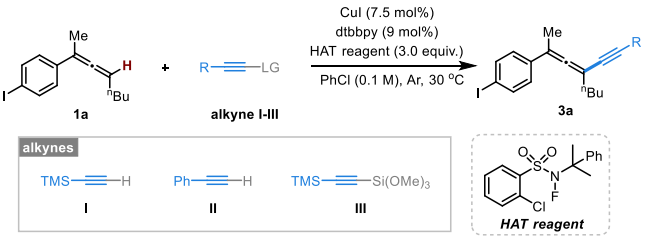
ently from the Liu group and our group, a radical strategy for the cyanation of the C(sp^2)−H bond of allenes has been reported.[11] This strategy involves the abstraction of a hydrogen atom from the sp^2-hybridized carbon atom by a reactive nitrogen-centered radical, resulting in the formation of a diffusible allenyl radical.

Motivated by these studies, we hypothesized that direct alkynylation of allene C−H bonds would represent an atom- and step-economic strategy for synthesizing allenynes under mild conditions, without the need for intricate substrates synthesis process or harsh reaction conditions. The greater efficiency of overlap between the π bond and the orbital of the single electron[12] (Figure 1D, right) suggests that the linear structure of the allenyl radical is thermodynamically more stable than the vinyl radical, resulting in the relatively lower BDE (~88 kcal/mol) of the allene C(sp^2)−H bond. However, several challenging issues must be addressed to realize allene C(sp^2)−H alkynylation: (1) Both allenes and alkynes in the starting materials and products are highly reactive in radical reactions due to their π system, making them prone to undergo radical addition to the C−C π bond; (2) Controlling site-selectivity during the hydrogen atom abstraction (HAA) process in the presence of both the allenyl C−H bond and other reactive C(sp^3)−H bonds is essential. Therefore, in this study, we utilize copper salt as a catalyst and *N*-fluoro-*N*-alkylsulfonamide as an HAT reagent[13] to achieve the direct alkynylation of allene C−H bonds under mild conditions. This approach yields good yields and exclusive selectivities (Figure 1D).

RESULTS AND DISCUSSION

A trisubstituted allene (**1a**) containing an allenyl C−H bond and two formal allyl C−H bonds, which could potentially compete in the hydrogen atom abstraction (HAA) process, was subjected to screening of reaction parameters (for more details, see Supporting Information). The initial experiment involved screening alkynylated reagents using CuI as the catalyst and 4,4′-di*tert*-butyl-2,2′-dipyridyl (dtbbpy) as the ligand. The reaction with trimethylsilylacetylene (**I**) yielded the desired allenyne (**3a**), albeit with a low yield (Table 1,

Table 1. Optimization of Reaction Conditions[a]



entry	variations of standard conditions	yield (%)
1	with the use of alkyne **I**	16
2	with the use of alkyne **II**	0
3	with the use of alkyne **III**	75
4	w/o Cu salt	0
5	w/o dtbbpy	0
6	w/o HAT reagent	0

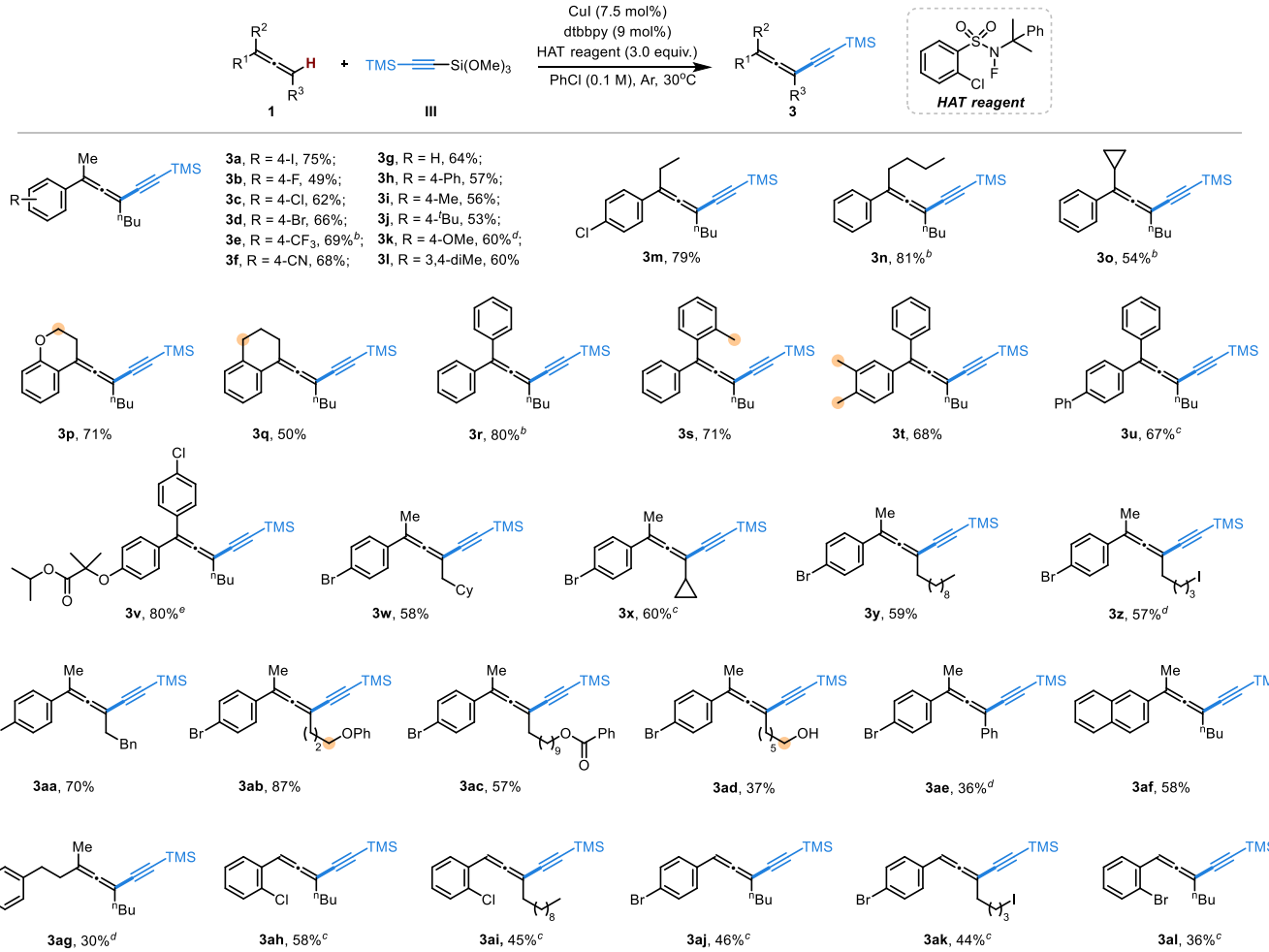
[a]Reaction conditions: **1** (0.2 mmol), alkyne **I**–**III** (0.8 mmol), CuI (7.5 mol %), dtbbpy (9 mol %), HAT reagent (0.6 mmol) in PhCl (0.1 M) under an Ar atmosphere, at 30 °C for 48 h.

Entry 1). However, when replacing alkyne (**I**) with phenylacetylene (**II**), the reaction failed to produce the corresponding product (Table 1, Entry 2). Interestingly, the reaction using disilyl-substituted acetylene (**III**) proceeded smoothly to yield the desired product (**3a**) with an excellent yield and perfect regioselectivity (Table 1, Entry 3), wherein C−Si cleavage occurred on the silyl ether ($Si(OMe)_3$) side rather than trimethylsilyl group (TMS). The selectivity of C−Si scission will be discussed in mechanistic studies.[14]

The optimized reaction conditions for regioselective alkynylation of allene C−H bonds were employed to investigate the generality of our protocol (Scheme 1). First, the substrate scope of trisubstituted allenes was examined. A variety of allenes bearing either electron-donating or electron-withdrawing aryl groups were found to be suitable substrates, affording the products (**3a**–**3l**) in moderate to excellent yields with good regioselectivities. Allene derivatives (**1m** or **1n**) with long pendant aliphatic chains also reacted efficiently to provide the corresponding alkynylated products (**3m** or **3n**). Additionally, allenes bearing with cyclopropyl, 3,4-dihydro-(1H)-benzopyranyl, and tetrahydronaphthyl groups were readily converted into the desired alkynylated compounds (**3o**–**3q**). Notably, in the case of **3o**, no "radical clock" process-driven ring-opening product derived from the propargylic radical, a resonance structure of the allenic radical, was detected. It is worth mentioning that the presence of ether (**3p**) or benzylic C−H bonds (**3q**) with low bond dissociation energies (BDE)

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ACS Catal. 2024, 14, 13219−13226

Scheme 1. Substrate Scope for sp^2 C−H Alkynylation[a]



[a]Reaction conditions: **1** (0.2 mmol), **2** (0.8 mmol), CuI (7.5 mol %), dtbbpy (9 mol %), N-F reagent (0.6 mmol) in PhCl (0.1 M) under an Ar atmosphere, at 30 °C for 48 h; [b]Replacing CuI with $CuBrCH_3SCH_3$ (7.5 mol %); [c]60 h; [d]70 h; [e]77 h; Yields of isolated products are given.

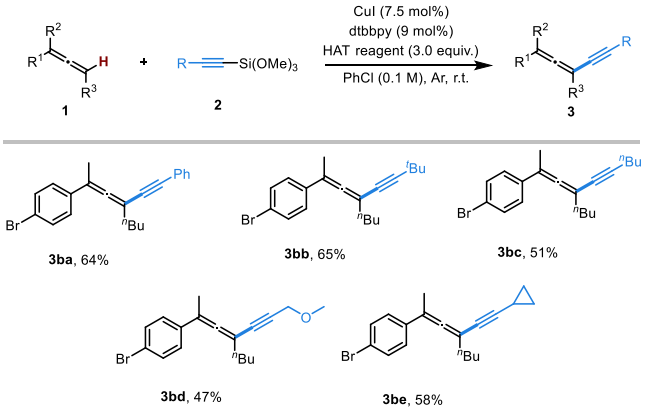
did not affect the site- selective hydrogen atom abstraction from the allenic C−H bond. Furthermore, 1,1-biaryl allenes were also compatible, yielding the desired products (**3r**−**3v**). This protocol exhibited excellent tolerance toward various structural units, including cyclohexane (**3w**), decane (**3y**), iodine (**3z**), and ester (**3ac**). Notably, the allenic C−H bond (**1x**) adjacent to a cyclopropyl ring underwent alkynylation readily without forming any ring-opening products via the "radical clock" process. Furthermore, reactive C(sp^3)−H bonds, such as those in benzyl group (**3aa**), phenyl ether (**3ab**), and unprotected alcohol (**3ad**), remained intact under these conditions. The reaction with 1,3-diaryl allene successfully yielded the desired product (**3ae**), albeit with a decreased yield likely due to high steric hindrance. Additionally, naphthyl-substituted were well-tolerated, leading to the formation of the corresponding product (**3af**) in synthetically useful yields. Remarkably, besides aryl-substituted allenes, allene substrate (**1ag**) bearing an alkyl group was also amenable to our protocol, affording the desired allenyne (**3ag**) with excellent regioselectivity. Next, the compatibility of disubstituted allenes with two allenic C(sp^2)−H bonds, which might compete in the HAA process (Scheme 1, bottom), was explored. Disubstituted allenes substituted with *ortho*- and *para*-halogenated aryl groups smoothly underwent site-specific HAT from the allenyl C−H bonds near the aliphatic chain with less steric hindrance. This led to a series of trisubstituted allenes (**3ah**−**3al**) with synthetically useful yields and excellent regioselectivities.

Subsequently, the scope of the alkynyl moiety of silyl ethers was examined with **1d** (Scheme 2). Both phenyl and alkyl alkynes could be readily incorporated into the allene scaffold, leading to corresponding allenynes (**3ba**−**3be**) with excellent regioselectivities. Of note, the highly reactive ether C(sp^3)−H bonds in both the alkynylated reagent (**2d**) and the product (**3bd**) remained intact in this HAT protocol.

The scaled-up preparation of **1a** proceeded readily and delivered a comparable yield, demonstrating the practicality of the protocol (Scheme 3A). The products can be conveniently transformed into other valuable compounds (Scheme 3B). For instance, treating the allene with *N*-bromosuccinimide (NBS) generated bromo-1,3-diene (**4**) through C−H bond elimination. Similar elimination occurred when **3a** was subjected to the reaction conditions with the use of $PhSO_2Na$ and an iron salt, yielding sulfonyl-1,3-diene (**5**). In the presence of KF, the TMS moiety in alkyne (**3a**) could be easily removed, and the desired product (**6**) underwent a click reaction with BnN_3 to afford *N*-heteroarene (**7**). As an important synthetic core, the alkynyl group in **3a** could be transformed into (*E*)-alkenyl

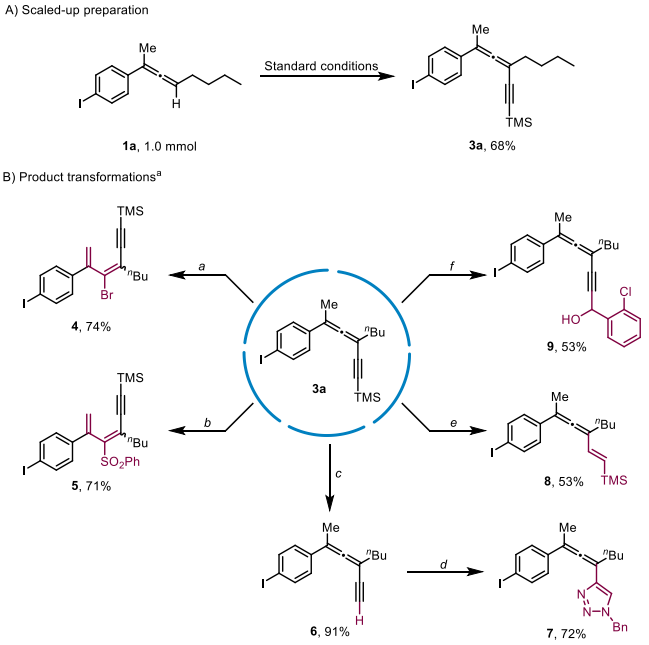
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Scheme 2. Substrate Scope for Alkynylating Reagent[a]



[a]Reaction conditions: 1 (0.2 mmol), 2 (0.8 mmol), CuI (7.5 mol %), dtbbpy (9 mol %), HAT reagent (0.6 mmol) in PhCl (0.1 M) under an Ar atmosphere, at 30 °C for 48 h.

Scheme 3. Scaled-Up Preparation and Product Transformations[a]

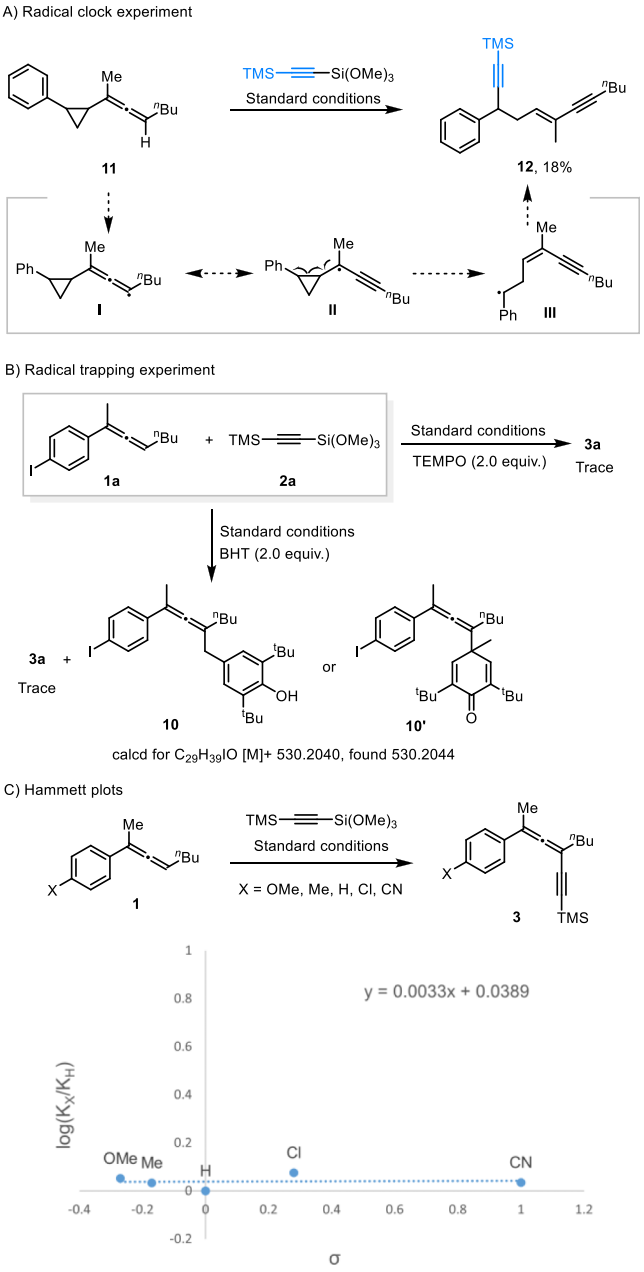


[a]Reaction condition: a) **3a** (0.1 mmol), NBS (1.3 equiv.), DCM (0.1 M), 30 °C, 13 h; b) **3a** (0.1 mmol), $PhSO_2Na$ (1.5 equiv.), $FeCl_3$•$6H_2O$ (2.5 equiv.), EtOH (0.2 M), 50 °C, 24 h; c) **3a** (0.1 mmol), KF (10 equiv.), EtOAc (0.5 ml), MeOH (0.5 ml), r.t., 36 h; d) **6** (0.09 mmol), BnN_3 (3 equiv.), $CuSO_4$•$5H_2O$ (50 mol%), sodium L-ascorbate (50 mol%), tBuOH (1 ml), H_2O (0.6 ml), N_2, rt, 3.5 h; e) **3a** (0.2 mmol), DIBAL-H (5 equiv.), Et_2O (0.1 M), 40 °C, 36 h; f): **3a** (0.1 mmol), 2-chlorobenzaldehyde (0.1 mmol), TBAF (10% in THF, 0.5 ml), THF (1 ml), 24 h; 1 M HCl (1 ml), 15 min.

allene (**8**) with high stereoselectivity. Nucleophilic addition of TMS-alkyne (**3a**) to an aldehyde resulted in the formation of a propargylic alcohol (**9**).

A radical-clock experiment was designed to detect the allenyl radical (Scheme 4A), and the desired ring-opening product (**12**) was obtained. These results confirmed the formation of the allenyl radical in the transformation. We also conducted radical trapping experiments using TEMPO and BHT (Scheme 4B). Under standard conditions, the addition of

Scheme 4. Mechanistic Studies



two equivalents of each trapping agent resulted in complete inhibition of the reaction. Additionally, possible trapping compounds, **10** and **10'**, were detected by mass spectrometry. Hammett plots suggest that the electronic properties of the aryl group on the allenes did not significantly influence the reaction efficiency, thereby potentially excluding an ionic reaction mode (Scheme 4C).

To rationalize the chemoselectivity, we conducted density functional theory (DFT) calculations to investigate the transmetalation between the Cu(II) intermediate **INT1** and the alkynylating reagent (**III**), with the results summarized in Figure 2A. Initially, the Cu(II) intermediate **INT1** forms a square pyramidal geometry complex with the alkynylating reagent (**III**), yielding the complex **INT2**. Subsequently, the alkynylating reagent (**III**) undergoes evolution into the transmetalated product **INT3** via an intermolecular nucleophilic attack of the alkynyl group at the Cu(II) center, leading

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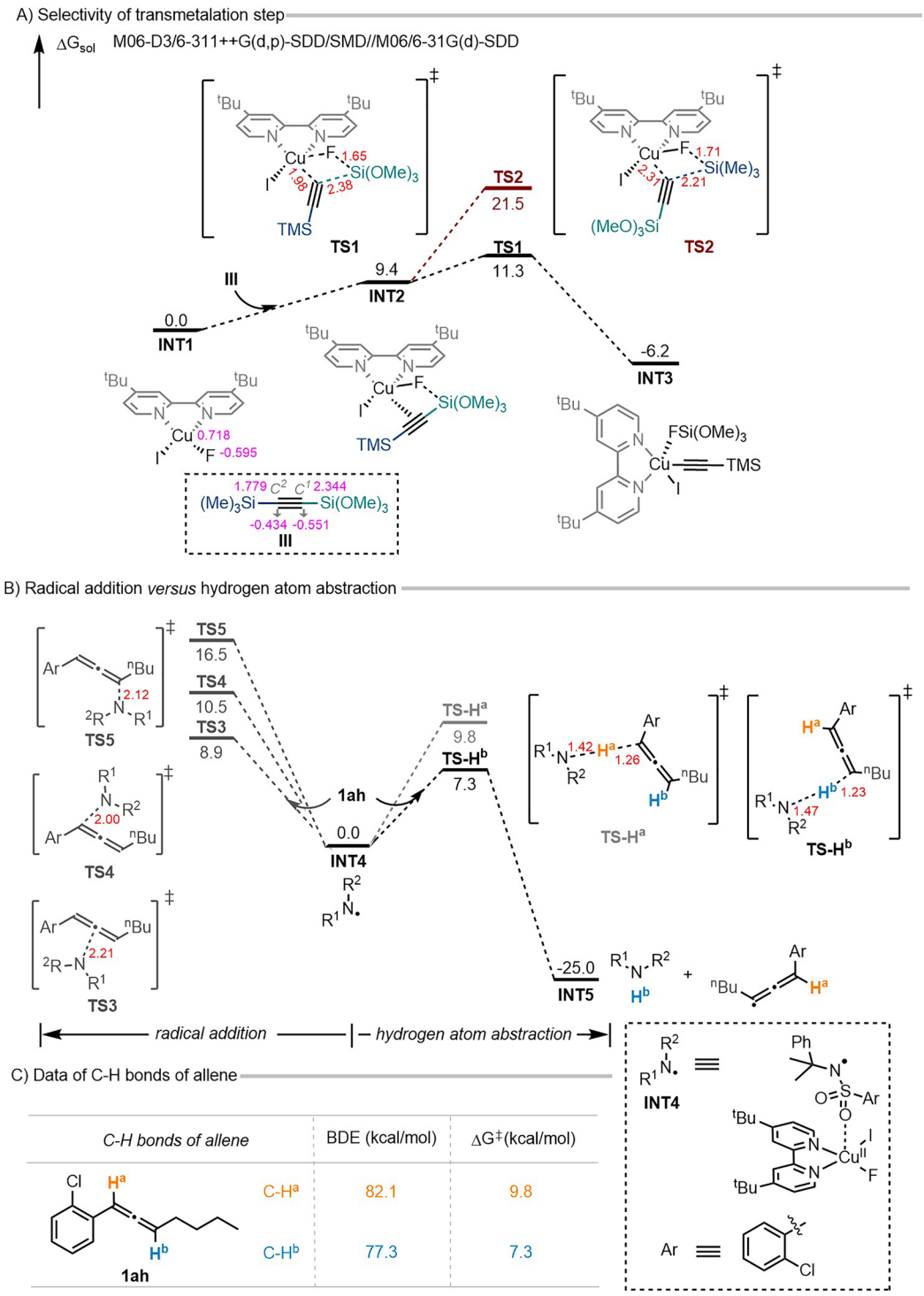


Figure 2. A) Energy profile (in kcal/mol) for the transmetalation step. The bond distances are given in Å (red color), and electron charges are in e (purple color). B) Radical addition versus hydrogen atom abstraction. C) Data of C−H bonds of allene.

to the release of $(CH_3O)_3SiF$, as indicated by TS1 with a relative energy of 11.3 kcal/mol. In contrast, for the release of $(CH_3)_3SiF$, the calculated $\Delta G^{\neq}$ for the transmetalation step via TS2 is 21.5 kcal/mol, which is 10.2 kcal/mol higher than TS1. This regioselectivity can be elucidated through NBO charge analyses for **III** and **INT1**. On one hand, in **III**, the C^1 atom exhibits a higher negative charge than the C^2 atom, −0.551 vs −0.434 *e*, favoring its nucleophilic attack on the Cu(II) atom. On the other hand, the calculated NBO charges on the F atoms of **INT1**, the Si atoms of $Si(OMe)_3$, and the Si atoms of $Si(Me)_3$ are −0.595, 2.344, and 1.779 *e*, respectively. This indicates that the electrostatic interaction between the F and Si atoms of $Si(OMe)_3$ is stronger than that between the F and Si atoms of $SiMe_3$, rationalizing the higher regioselectivity of the transmetalation step.

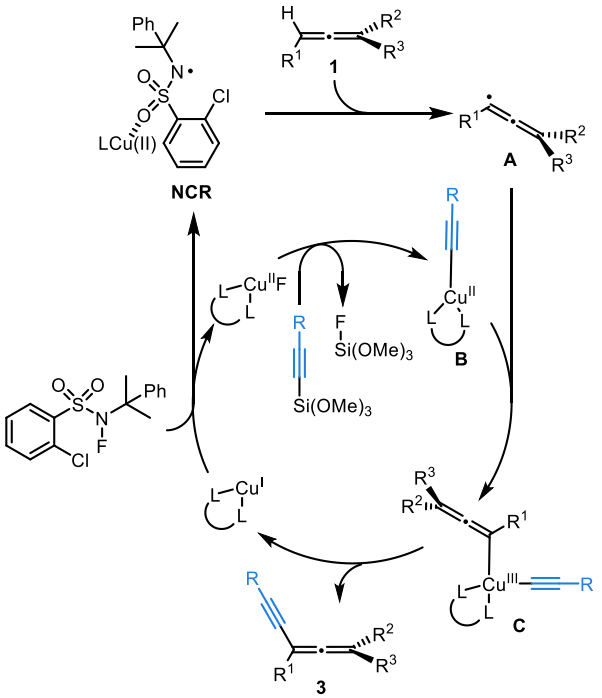
To gain a deeper understanding of the chemo- and regioselectivity involved in the alkynylation of the allenyl $C(sp^2)$−H bond, DFT calculations were conducted (Figure 2B,C). For

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ACS Catal. 2024, 14, 13219−13226

instance, considering compound (**1ah**) with two allenyl C(sp^2)−H bonds, the activation energy ($\Delta G^{\neq}$) for the hydrogen atom abstraction (HAA) process involving these C−H bonds and the ΔG of this step were computed and depicted in Figure 2B. The results illustrate that the HAA from the allenyl C(sp^2)−H^b bond exhibits the lowest activation energy (7.3 kcal/mol) and is exergonic by 25.0 kcal/mol. In contrast to the HAA process, investigations into the addition of the *N*-centered radical to the allene π bonds reveal relatively higher activation barriers (Figure 2B, left). Furthermore, to explore the regioselectivities of these reactions, additional data such as the bond dissociation energies (BDEs) of the C(sp^2)−H bonds of substrate (**1ah**) and the activation energy ($\Delta G^{\neq}$) for the HAA process were calculated and depicted in Figure 2C. These findings demonstrate that allenyl C(sp^2)−H bonds, characterized by lower bond dissociation energies (BDEs), are predisposed to facile alkynylation through the HAA process with relatively lower activation barriers. Importantly, the BDEs of both the allenyl C(sp^2)−H^a and C(sp^2)−H^b bonds are even lower than those of certain activated C(sp^3)−H bonds, such as those found in toluene (89.7 kcal/mol), allylic C−H bonds (approximately 88 kcal/mol), ethanol (95.9 kcal/mol), tetrahydrofuran (THF, 92.1 kcal/mol), among others. This observation elucidates the exclusive selectivity observed at the allenyl C(sp^2)−H bond, even in the presence of some activated C(sp^3)−H bonds.

Based on previous reports and mechanistic studies, we propose a plausible pathway (Scheme 5). The interaction between Cu(I) and the HAT reagent forms a Cu(II)-bound N-centered radical (**NCR**). This Cu-complex-bound **NCR** species selectively abstracts a hydrogen atom from the allene C(sp^2)−H bond to generate the allenyl radical (**A**).[15] The interaction between the Cu(II)F complex and the alkynylating reagent yields a Cu(II)-alkyne species (**B**), which, upon interception by **A**, forms **C**. The subsequent reductive elimination of **C** yields the final product (**3**) and regenerates Cu(I) to continue the catalytic cycle.

Scheme 5. Proposed Mechanism



CONCLUSION

In summary, we have presented a novel copper-catalyzed, site-selective radical alkynylation of allenyl C(sp^2)−H bonds under mild conditions. This method affords a range of valuable functionalized allenyne derivatives in synthetically useful yields. DFT calculations offer theoretical insights into chemo-/regioselectivity of HAT process and the crucial role of the silylether moiety as a removal group in the alkynylating reagent. The approach demonstrates a broader substrate scope, tolerance toward various functional groups, and yields high product diversity.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.4c03688.

Experimental procedures and characterization data (PDF)

AUTHOR INFORMATION

Corresponding Authors

Jiang-Kai Qiu − *College of Biotechnology and Pharmaceutical Engineering, Nanjing Tech University, Nanjing 211816, China*; Email: qiujiangkai@njtech.edu.cn

Kai Guo − *College of Biotechnology and Pharmaceutical Engineering, Nanjing Tech University, Nanjing 211816, China*; orcid.org/0000-0002-0013-3263; Email: guok@njtech.edu.cn

Xiaoguang Bao − *Key Laboratory of Organic Synthesis of Jiangsu Province, College of Chemistry, Chemical Engineering and Materials Science, Soochow University, Suzhou, Jiangsu 215123, China*; Email: xgbao@suda.edu.cn

Xinxin Wu − *Key Laboratory of Organic Synthesis of Jiangsu Province, College of Chemistry, Chemical Engineering and Materials Science, Soochow University, Suzhou, Jiangsu 215123, China*; orcid.org/0000-0003-4603-7514; Email: xxwu99@suda.edu.cn

Authors

Xiu Duan − *College of Biotechnology and Pharmaceutical Engineering, Nanjing Tech University, Nanjing 211816, China; Key Laboratory of Organic Synthesis of Jiangsu Province, College of Chemistry, Chemical Engineering and Materials Science, Soochow University, Suzhou, Jiangsu 215123, China*

Qihao Jin − *Key Laboratory of Organic Synthesis of Jiangsu Province, College of Chemistry, Chemical Engineering and Materials Science, Soochow University, Suzhou, Jiangsu 215123, China*

Kaifeng Wang − *Key Laboratory of Organic Synthesis of Jiangsu Province, College of Chemistry, Chemical Engineering and Materials Science, Soochow University, Suzhou, Jiangsu 215123, China*

Yifan Sun − *Key Laboratory of Organic Synthesis of Jiangsu Province, College of Chemistry, Chemical Engineering and Materials Science, Soochow University, Suzhou, Jiangsu 215123, China*

Youhao Wei − *Key Laboratory of Organic Synthesis of Jiangsu Province, College of Chemistry, Chemical Engineering and*

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Materials Science, Soochow University, Suzhou, Jiangsu 215123, China

Zheng Wang − *Key Laboratory of Organic Synthesis of Jiangsu Province, College of Chemistry, Chemical Engineering and Materials Science, Soochow University, Suzhou, Jiangsu 215123, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acscatal.4c03688

Author Contributions

#X.D., Q.J., and K.W. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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